pfi

Python for Fixed Income

COURSE 6 · NEURAL NETWORKS

Course 6 Capstone

WEEK 3 · DAY 5

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This course is educational content created in a personal capacity. Nothing here is investment advice or a recommendation to buy or sell any security. All portfolio data is synthetic.

The scenario: does a network do better?

The colleague from Course 5's capstone comes back. You chose LightGBM on the Taiwan file.

On the same rows, by the same folds, does a network do better?

And how sure can anyone be?

Nothing in the notebook promises which model wins. The checks were written after the numbers were computed.

What the answer is not

Not a credit view on any company

Not a rating

Not a probability going forward

Not advice to a lender or an investor

It is a description of a historical, anonymised dataset: what two models show on it, and where they are wrong.

What you hand in: twelve deliverables

#	Deliverable
1	netkit complete, pytest passing
2	course5_training_rows(path): training rows only
3	The rows described, hashed checks
4	build_network, network_transformer, class weights
5	build_lightgbm, by the same folds
6	Paired folds, fold seeds, intervals

#	Deliverable
7	Thresholds for recall 0.70, flags, precision
8	A final choice with its reasons
9	A results workbook, read back
10	What the model shows, and where it fails
11	One commit per file, then git tag course6
12	An AI-use log

The functions go into one module of your own, capstone.py, which grows question by question.

The rows: closed means closed

Course 5's training rows

Fitted and scored by five folds. Every setting, threshold, and choice is made here.

Course 5's test rows

Opened once in Course 5. Never scored again.

```
X_train, y_train = capstone.course5_training_rows(DATA_FILE)   # the training rows, and only those
```

Three fold seeds

The comparison runs on folds shuffled with seeds 42, 43, 44. A gap that holds on all three says more than one set of folds can.

What makes it hard: nothing is planted

Two scales

small ratios in most rows, billions in others

A rare outcome

a few dozen bankrupt rows per fold

A closed test set

Course 5 opened it; folds only

A network's settings

width, depth, dropout, transformer, weights

The size regime

same digits on one thread and every core

The comparison

every gap read beside its noise

Trees split on a column's order. A network multiplies every input by a weight, so it notices scale.

Nine sections, three sittings

First sitting

1 Your module

2 The training rows

3 The rows described

4 The network

Rows, then the network

Second sitting

5 LightGBM

6 The comparison

7 Thresholds

Compare, honestly

Third sitting

8 Choice and workbook

9 Description and log

Choose and describe

A five-fold network run takes about half a minute on one thread; section 4 runs five, section 6 two more. LightGBM takes seconds.

The network, one setting at a time

The start: 32 and 16 ReLU units

First layer's size, batch x inputs x units: 64 x 92 x 64 is 376,832, at the edge of the regime; 64 x 92 x 32 is 188,416, well inside.

Four rungs, one change each

Model 1: the starting network

Model 2: the transformer

Model 3: dropout

Model 4: the depth

The rule for keeping a change, written before any fold is scored

a change is kept when its fold mean average precision is higher than the network kept so far and it is higher on at least 3 of the 5 folds. Otherwise the network kept so far stays.

Class weights: two numbers decide

Fold mean average precision

The ladder's number, by the same five folds.

Precision at the desk's recall

Threshold for recall 0.70 on out-of-fold outputs, then how pure the flagged list is.

Kept only if both are higher. If they disagree, the simpler model stays.

Allowed names

the model's score (weighted)

output versus rate, measured

Never called

a probability (weighted)

a credit view on any company

LightGBM, by the same folds

```
LGBMClassifier(n_estimators=300, learning_rate=0.05, num_leaves=8, random_state=SEED, n_jobs=1, verbose=-1)
```

```
lightgbm_oof = cross_val_predict(capstone.build_lightgbm(), X_train, y_train, cv=mlkit.cv_folds(),
                                 method="predict_proba", n_jobs=1)[:, 1]
```

The same folds

mlkit.cv_folds() is the splitter cv_network uses, so each fold's two scores come from the same held-out rows.

No transformer

Trees split on each column's order, so the two scales cost LightGBM nothing.

An honest comparison: three tools

Paired fold differences

Network less LightGBM on each fold's held-out rows. Read the mean beside the fold standard deviation.

Three fold seeds

The same comparison on two more shuffles. What moves between seeds is the noise of the folds.

A paired bootstrap

bootstrap_interval's own 1,000 resamples, both models scored on each. The noise from which rows were scored.

Two overlapping intervals settle nothing. The paired interval asks the paired question.

Thresholds, and the choice

Threshold: recall of at least 0.70

```
mlkit.threshold_for_recall(
    y_train, network_oof, TARGET_RECALL)
```

On out-of-fold outputs of the training rows. Then both models side by side: rows flagged and precision for the same recall. An illustration, not a lending rule.

The choice, with its reasons

The primary metric on the folds

The paired interval

What the fold seeds showed

Flags at the desk's recall

Run time, said in words

How much numbers move between runs

What the desk asked

How the checks work

A check holds a short code, never the answer

```
check('Question 3.1', answer_3_1, 'a short code', places=2)
```

```
correct
```

```
not yet. You have ...
```

```
not attempted yet
```

Store the full number; check rounds to the places asked. Several values in a tuple.

Decisions change the numbers

Checks after section 4 assume the course's decisions. "Stuck after trying?" blocks hold a decision, never a count: make your own first.

33 questions and 29 check lines, in the order an analyst works.

The chart cells: provided, and shy until you answer

#	Chart cell
1	The four configurations, by fold
2	The kept network, with and without class weights
3	The two models, fold by fold
4	The three sets of folds
5	Pooled values with intervals, and the paired resamples
6	Precision-recall and ROC, each threshold marked
7	Confusion matrices at the thresholds

7 chart cells. The notebook shows no result before you have worked for it.

Not attempted yet

Each cell checks that your variables exist. Until they do, it prints not attempted yet and draws nothing. Once they do, it draws your results, with a styled table where it compares.

The helper answered(...) is true when no name it needs is still the ... placeholder.

The workbook, the description, Git

The workbook

Six sheets, read back. Excel rebuilds fold means, paired differences, flags, counts, and the paired interval.

The description

300 to 500 words, checked by netkit.assert_history: no forecast, cause, or later-date words.

Git

One commit per deliverable that writes a file, then git tag course6.

Allowed names

model probability of bankruptcy in this file

Never called

a credit view on any company

The AI check

The docstring is the prompt. The code that comes back runs, and its number looks fine.

1

Docstring

Which rows fit, which rows stop, which rows report: what you asked for.

→

2

Code back

Written for the course in the style of an AI answer. Read it.

→

3

Test first

Assert what the docstring promises. Run it: it fails.

→

4

Fix

Find the rule it broke, fix it, run the test again.

Today: a fold loop that runs and returns an output for every row. Check where its stopping rows come from.

Code that runs is not the same as code that is right.

What you can do now

- Compute a gradient by hand and check it against TensorFlow
- Build, fit, and read a Keras network line by line
- Change one setting at a time; say how much is the seed
- Score sequence models walk-forward, and say what they did
- Compare a network with boosting, beside the noise

Next keep your my_bondmath folder up to date, and bring it with you.